Round 11 — Change What is Represented

Corrected technical assessment of the aggregated rank-k old third-cumulant proposal

Bottom line. The supplied aggregated rank-k idea is a stronger candidate than the earlier query-only construction because it preserves long-lived old third-cumulant information in a transport-compatible representation. However, the previous PDF overstated what follows from a rank-230 Frobenius approximation and understated the cost/accounting and nonlinear recompression issues. The scheme should be treated as a promising candidate, not as a demonstrated ≤125u solution.

1. State carried per layer

Carry the usual mean μ and covariance C, plus an aggregate old third-cumulant representation:

$$T_{old} \approx \Sigma_{a=1..k} \sigma_a \cdot u_a \otimes u_a \otimes v_a.$$

This removes per-source dense legs and never materializes the full n×n×n tensor. The representation is transport-compatible under a linear map B because each factor can be pushed forward:

$$u_a \leftarrow B\,u_a,\quad v_a \leftarrow B\,v_a.$$

2. What is genuinely good about it

- It attacks the right object. The old κ3 contribution is long-lived and coherent, while the final mean is mainly sensitive through next-layer mean shifts. An aggregate representation can preserve that memory without retaining every source separately.
- Transport is exact before truncation. If the current aggregate representation is exact, linear transport of each rank-one component is exact. The approximation enters through the initial truncation and any later recompression, not through the transport itself.
- It is directly falsifiable. The stored teacher arrays can test the exact channels that matter: variance, diagonal d3, K4d, and the covariance entries feeding them.
- It is plausibly outside the measured dead list. It is not source-by-source storage, and it does not require a shared basis for all individual source legs. It is an aggregate representation of the combined old contribution.

3. The important corrections

A. Rank-230 D21 error does not imply 2% channel error. A Frobenius-optimal rank-k approximation minimizes $||D21-D21_k||_F$, but the next-layer variance uses specific W-weighted quadratic queries. A small global Frobenius residual can still concentrate in directions that matter strongly to those queries. Therefore the decisive quantities are the actual channel errors, not D21 Frobenius error alone.

B. The d3 factor needs a convention check. If $T = \Sigma\, \sigma_a\, u_a \otimes u_a \otimes v_a$ literally, then $T_{iii} = \Sigma\, \sigma_a\, u_{ai}^2\, v_{ai}$. A factor of 3 is not automatic. It is valid only if the stored representation is a specifically symmetrized convention whose diagonal expands with that factor. Match this exactly to the teacher definition before testing.

C. The nonlinear/ReLU recompression step is the key missing algorithmic detail. Linear transport can preserve the factors, but after the ReLU/young-source join the aggregate old contribution must either remain representable without rank growth or be deterministically recompressed. If recompression requires forming the dense tensor, the claimed cost advantage disappears. The implementation therefore needs an explicit factor-space update/recompression rule and its cost must be included.

D. The quoted cost needs normalization. For n=1024 and k=230, two n×n matvecs per component require about 4.82×10^8 multiply/add-scale operations per layer. Relative to n^3 this is approximately $0.449\, n^3$, not $0.22\, n^3$. The ~400-neuron contraction is about 9.42×10^7 operations, approximately $0.090\, n^3$. Use the competition's exact unit convention when converting these counts to u. In any

case, the old-tier overhead is not enough by itself to establish a total ≤125u if the young tier remains near 153u.

4. Correct channel formulas

For $T_{ijk} \approx \Sigma \sigma_a u_{ai} u_{aj} v_{ak}$, the required D21 slice is:

$$D21_{i,c} = \Sigma_a \sigma_a \cdot u_{ai}^2 \cdot v_{ac}.$$

The old contribution to the variance of output neuron i is:

$$\delta Var_i = \Sigma_a \sigma_a \cdot [w_i^T(u_a \circ u_a)] \cdot [w_i^T v_a].$$

The diagonal third cumulant must follow the exact teacher convention; under the literal unsymmetrized tensor above:

$$d3_i = \Sigma_a \sigma_a \cdot u_{ai}^2 \cdot v_{ai}.$$

5. The decisive Stage-1 screen

Do not use D21 Frobenius error as the main gate. Use the stored teacher arrays to compare the actual mean-relevant channels. At layers 7, 11, 14, 15 and 16, report relative errors separately for diagonal and off-diagonal quantities.

Channel	Required test
Variance	Relative error of the old-source variance correction; emphasize the W-weighted query.
d3	Relative error per neuron using the exact teacher convention.
K4d	Relative error in the diagonal fourth cumulant after the closure.
Covariance	Diagonal and off-diagonal C errors at the layers used by the chain.

Use the user's gate exactly: kill the method if any channel is ≥1.5× the chain error at two or more deep layers, or if its error grows with depth faster than the chain. Also perform both teacher-forced one-step tests and a true free-run from layer 1; the latter is essential because a source-free or locally accurate approximation can accumulate memory error.

6. Comparison with the earlier alternative

Criterion	Earlier query-focused idea	Aggregated rank-k proposal
Object represented	Only mean-relevant projections	Aggregate old κ3 / D21 structure
Long-memory preservation	Potentially incomplete	Explicitly retained
Linear transport	Query-dependent	Exact on retained factors
Source legs	Avoided	Avoided
Main risk	Missing an important query	Rank growth / recompression + query mismatch
Falsifiability	Moderate	Strong with stored teacher arrays
Current status	More speculative	More promising, still unproven

7. Recommended single falsifier

At layer 7, take the exact old-source aggregate from the teacher data. Construct its best rank-k representation for k = 100, 150, 200, 230, 300. For each k, immediately evaluate the three quantities that matter to the next-layer means: (i) the W-weighted variance correction for the active neurons, (ii) d3, and (iii) K4d after the existing closure. Then run the same factor state forward without teacher resets to layers 11, 14, 15 and 16. This single experiment simultaneously tests compression quality, query alignment, and memory stability.

Interpretation: If k=230 passes the channel gate and the free-run errors remain within the chain's gate, the representation is worth integrating. If it fails, the failure mode tells you what to fix: a variance-only failure indicates query-misaligned compression; a depth-growing failure indicates rank loss/recompression error; a d3/K4d failure indicates that D21 alone is not sufficient to reconstruct the channels required by the closure.

8. Stage 2 target

Only after Stage 1 passes should the method be integrated into the chain. Measure final-layer mean MSE against the 1e9-sample truth, paired with the chain, and then test across 100 MLPs on Kaggle. Apply the supplied score thresholds exactly: MSE ≤ 1.07× chain at 102u, ≤ 0.95× at 115u, or ≤ 0.84× at 125u. A prototype is plausibly useful at ≤0.8× chain score, but that is not the final target.

Final assessment

Verdict: the aggregated rank-k mechanism is the better Round-11 direction. Its central insight—compress the aggregate long-memory third-cumulant content rather than each source leg—is exactly the representation change the measurements suggest. But the claim that k=230 gives the required channel accuracy is not established by a 2% Frobenius D21 result, and the total-cost claim must be recomputed using the same u normalization as the chain. The next experiment should therefore be a channel-aware, free-run rank sweep, not another argument from matrix rank alone.

Note: This corrected version intentionally avoids the broken LaTeX rendering that affected the previous PDF. Equations are rendered as plain Unicode text so they remain legible in standard PDF viewers.